

Package ‘manMetaVAR’

November 24, 2025

Title Multivariate Meta-Analysis of Vector Autoregressive Model
Coefficients

Version 0.9.6

Description Research compendium for the manuscript
Pesigan, I. J. A., et al. (In Preparation).
Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients.
<doi:10.0000/0000000000>.

URL <https://github.com/jeksterslab/manMetaVAR>,
<https://jeksterslab.github.io/manMetaVAR/>,
<https://osf.io/uenr7/>, <https://doi.org/10.0000/0000000000>

BugReports <https://github.com/jeksterslab/manMetaVAR/issues>

License MIT + file LICENSE

Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

Depends R (>= 4.0.0), OpenMx, fitDTVARMxID, metaVAR

Imports stats, utils, simStateSpace, mlVAR, ggplot2

Remotes jeksterslab/fitDTVARMxID, jeksterslab/metaVAR

Suggests knitr, rmarkdown, testthat

RoxygenNote 7.3.3.9000

NeedsCompilation no

Author Ivan Jacob Agaloos Pesigan [aut, cre, cph] (ORCID:
<<https://orcid.org/0000-0003-4818-8420>>)

Maintainer Ivan Jacob Agaloos Pesigan <r.jeksterslab@gmail.com>

Contents

Compress	2
Dynamics	3

FigBias	4
FigCoverage	5
FigPower	6
FigRMSE	7
FigType1Error	8
FitDTVAR	9
FitMetaVAR	10
FitMLVAR	11
FitMplus	11
GenData	13
manmetavar-data-methods	13
manmetavar-dtvar-methods	14
manmetavar-metavar-methods	16
manmetavar-mlvar-methods	17
manmetavar-mplus-methods	18
MplusInput	19
params	21
results	21
Sim	22
SimFitDTVAR	24
SimFitMetaVAR	25
SimFitMLVAR	26
SimFitMplus	27
SimFN	28
SimGenData	28
SimProj	29
Sum	30
SumFitMetaVARLB	31
SumFitMetaVARNormal	31
SumFitMLVAR	32
SumFitMplus	33
Index	34

Compress

Compress Replication

Description

Compress Replication

Usage

Compress(taskid, repid, output_folder)

Arguments

- taskid Positive integer. Task ID.
- repid Positive integer. Replication ID.
- output_folder Character string. Output folder.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

Dynamics	<i>Process Dynamics</i>
----------	-------------------------

Description

Process Dynamics

Usage

Dynamics(dynamics)

Arguments

- dynamics 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

See Also

Other Data Generation Functions: [GenData\(\)](#)

Examples

```
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)
```

FigBias

*Plot Relative Bias***Description**

Plot relative bias for the fixed and random effects.

Usage

```
FigBias(
  results,
  bias = TRUE,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  rm_zero = FALSE,
  ylim = c(-0.15, 0.15)
)
```

Arguments

results	results data frame.
bias	Logical. If bias = TRUE, plot absolute bias. If bias = FALSE, plot relative bias. Note that when parameter value is equal to zero, absolute bias is used instead of relative bias.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
rm_zero	Logical. Remove cases where the population parameter is equal to zero.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigCoverage\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#), [FigType1Error\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigBias(results)

## End(Not run)
```

FigCoverage

Plot Coverage Probabilities

Description

Plot coverage probabilities for the fixed and random effects.

Usage

```
FigCoverage(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  rm_zero = FALSE,
  ylim = c(0, 1)
)
```

Arguments

results	results data frame.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
rm_zero	Logical. Remove cases where the population parameter is equal to zero.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#), [FigType1Error\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigCoverage(results)

## End(Not run)
```

FigPower

*Plot Statistical Power***Description**

Plot statistical power probabilities for the fixed and random effects.

Usage

```
FigPower(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  rm_zero = FALSE,
  ylim = c(0, 1)
)
```

Arguments

results	results data frame.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
rm_zero	Logical. Remove cases where the population parameter is equal to zero.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigRMSE\(\)](#), [FigType1Error\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigPower(results)

## End(Not run)
```

FigRMSE*Plot RMSE*

Description

Plot RMSE for the fixed and random effects.

Usage

```
FigRMSE(  
  results,  
  dynamics = 1,  
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),  
  rm_zero = FALSE,  
  ylim = c(0, 0.25)  
)
```

Arguments

results	results data frame.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
rm_zero	Logical. Remove cases where the population parameter is equal to zero.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigPower\(\)](#), [FigType1Error\(\)](#)

Examples

```
## Not run:  
data(results, package = "manMetaVAR")  
FigRMSE(results)  
  
## End(Not run)
```

FigType1Error

Plot Type 1 Error Rates

Description

Plot type 1 error rates for zero random effects.

Usage

```
FigType1Error(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  ylim = c(0, 0.25)
)
```

Arguments

results	results data frame.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigType1Error(results)

## End(Not run)
```

FitDTVAR*Fit the Model using the fitDTVARmxD Package*

Description

The function fits the model using the [fitDTVARmxD::fitDTVARmxD](#) package.

Usage

```
FitDTVAR(data, wd = ".", ncores = NULL, seed = NULL)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>wd</code>	Character string. Working directory.
<code>ncores</code>	Positive integer. Number of cores to use.
<code>seed</code>	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

Description

The function performs multivariate meta-analysis using the [metaVAR::metaVAR](#) package.

Usage

```
FitMetaVAR(data, fit, ncores = NULL)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>fit</code>	R object. Output of the FitDTVAR() function.
<code>ncores</code>	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  data = data,
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```

FitMLVAR*Fit the Model using the mlVAR Package*

Description

The function fits the model using the [mlVAR::mlVAR](#) package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:  
set.seed(42)  
data <- GenData(taskid = 1)  
fit <- FitMLVAR(data = data)  
print(fit)  
summary(fit)  
  
## End(Not run)
```

FitMplus*Fit the Model using Mplus*

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 120000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL
)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>chains</code>	Integer. Number of chains.
<code>iter</code>	Integer. Number of iterations.
<code>fscores</code>	Integer. Number of iterations for factor scores.
<code>plot</code>	Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3;</code> to Mplus input file.
<code>default_priors</code>	Logical. If <code>default_priors = TRUE</code> , use default priors.
<code>wd</code>	Character string. Working directory.
<code>mplus_bin</code>	Character string. Path to Mplus binary. If <code>mplus_bin = NULL</code> , the function will try to find the appropriate binary.
<code>ncores</code>	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

GenData

*Simulate Data***Description**

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid)
```

Arguments

`taskid` Positive integer. Task ID.

See Also

Other Data Generation Functions: [Dynamics\(\)](#)

Examples

```
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)
```

manmetavar-data-methods

*Methods for Objects of Class manmetavar.data***Description**

This page documents the available methods for objects of class `manmetavar.data`.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

x	Object of class <code>manmetavar.data</code> .
...	additional arguments.
object	Object of class <code>manmetavar.data</code> .

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-dtvar-methods

Methods for Objects of Class `manmetavar.dtvvar`

Description

This page documents the available methods for objects of class `manmetavar.dtvvar`.

Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
```

```

    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

```

Arguments

object	Object of class <code>manmetavar.dtvar</code> .
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.

nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
...	additional arguments.
x	Object of class <code>manmetavar.dtvvar</code> .
means	Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.
digits	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-metavar-methods

Methods for Objects of Class `manmetavar.metavar`

Description

This page documents the available methods for objects of class `manmetavar.metavar`.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)
```

```
## S3 method for class 'manmetavar.metavar'
vcov(object, ...)
```

```
## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)
```



```
## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)
```

Arguments

object	Object of class <code>manmetavar.metavar</code> .
...	additional arguments.
x	Object of class <code>manmetavar.metavar</code> .
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
lb	Logical. If TRUE, returns profile likelihood-based confidence intervals. If FALSE, returns Wald confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-mlvar-methods

Methods for Objects of Class `manmetavar.mlvar`

Description

This page documents the available methods for objects of class `manmetavar.mlvar`.

Usage

```
## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mlvar'
summary(
```

```

    object,
    show = c("fit", "temporal", "contemporaneous", "between"),
    digits = 4,
    ...
  )

```

Arguments

x	Object of class <code>manmetavar.mlvar</code> .
show	Tables to show.
digits	Integer indicating the number of decimal places to display.
...	additional arguments.
object	Object of class <code>manmetavar.mlvar</code> .

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-mplus-methods

Methods for Objects of Class `manmetavar.mplus`

Description

This page documents the available methods for objects of class `manmetavar.mplus`.

Usage

```

## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,

```

```

    burnin = NULL,
    legend_loc = "topright",
    ...
)

```

Arguments

object	Object of class <code>manmetavar.mplus</code> .
median	Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.
burnin	Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.
...	additional arguments.
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
x	Object of class <code>manmetavar.mplus</code> .
what	Character string. If <code>what = "posterior"</code> , return posterior distribution plots. If <code>what = "trace"</code> , return trace plots.
legend_loc	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

MplusInput	<i>Create Mplus Input File</i>
------------	--------------------------------

Description

The function creates an Mplus input file.

Usage

```

MplusInput(
  dynamics,
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,

```

```

    fscores,
    plot,
    default_priors,
    ncores = NULL
)

```

Arguments

dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use default priors.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#)

Examples

```

cat(
  MplusInput(
    dynamics = 1,
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 60000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = TRUE,
    ncores = NULL
  )
)

```

params	<i>Simulation Parameters</i>
--------	------------------------------

Description

Simulation Parameters

Usage

data(params)

Format

A dataframe with 27 rows and 4 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occasions.

dynamics Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

Author(s)

Ivan Jacob Agaloos Pesigan

results	<i>Simulation Results</i>
---------	---------------------------

Description

Simulation Results

Usage

data(results)

Format

A dataframe with I rows and J columns:

taskid Simulation Task ID.

replications Number of replications.

parnames Parameter names.

parameter Population parameter value.

method Method used to fit the data.

n Sample size.

time Number of measurement occasions.

dynamics Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

ci Confidence/credible intervals method.

est Mean parameter estimate.

se Mean standard error.

t Mean test statistic.

p Mean p -value.

ll Mean lower limit of the 95% confidence/credible interval.

ul Mean upper limit of the 95% confidence/credible interval.

sig Proportion of statistically significant results.

theta_hit Proportion of replications where the confidence intervals contained the population parameter.

sq_error Mean squared error.

bias Bias.

rel_bias Relative bias.

se_bias Bias in standard error estimate.

rel_se_bias Relative bias in standard error estimate.

rmse Root mean square error.

coverage Coverage probability.

power Statistical power.

par_idx Parameter index.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim

Simulation Replication

Description

Simulation Replication

Usage

```

Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  metavar,
  mlvar,
  mplus,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)

```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
data	Logical. Simulate data.
metavar	Logical. If metavar = TRUE, fit the model using the metaVAR package.
mlvar	Logical. If mlvar = TRUE, fit the model using the mlVAR package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use default priors.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR

Simulation Replication - FitDTVAR

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR	<i>Simulation Replication - FitMetaVAR</i>
---------------	--

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMLVAR

Simulation Replication - FitMLVAR

Description

Simulation Replication - FitMLVAR

Usage

```
SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus

*Simulation Replication - FitMplus***Description**

Simulation Replication - FitMplus

Usage

```

SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)

```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3</code> ; to Mplus input file.
default_priors	Logical. If <code>default_priors = TRUE</code> , use default priors.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
-------	-----------------------------

Description

Simulation File Name

Usage

SimFN(output_type, output_folder, suffix)

Arguments

- output_type Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var".
- output_folder Character string. Output folder.
- suffix Character string. Output of manCTMed::SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
------------	---

Description

Simulation Replication - GenData

Usage

SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj

Simulation Project Name

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

Sum	Summary
-----	---------

Description

Summary

Usage

```
Sum(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  metavar,
  mlvar,
  mplus,
  ncores
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
metavar	Logical. If metavar = TRUE, fit the model using the metaVAR package.
mlvar	Logical. If mlvar = TRUE, fit the model using the mlVAR package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
ncores	Positive integer. Number of cores to use.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARLB	<i>Summary: Likelihood-Based (FitMetaVAR)</i>
-----------------	---

Description

Summary: Likelihood-Based (FitMetaVAR)

Usage

```
SumFitMetaVARLB(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal	<i>Summary: Normal Theory (FitMetaVAR)</i>
---------------------	--

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMLVAR

Summary (FitMLVAR)

Description

Summary (FitMLVAR)

Usage

```
SumFitMLVAR(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplus	<i>Summary (FitMplus)</i>
-------------	---------------------------

Description

Summary (FitMplus)

Usage

SumFitMplus(taskid, reps, output_folder, overwrite, integrity, ncores)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

Index

* **Compression Functions**

Compress, [2](#)

* **Data Generation Functions**

Dynamics, [3](#)

GenData, [13](#)

* **Figure Functions**

FigBias, [4](#)

FigCoverage, [5](#)

FigPower, [6](#)

FigRMSE, [7](#)

FigType1Error, [8](#)

* **Model Fitting Functions**

FitDTVAR, [9](#)

FitMetaVAR, [10](#)

FitMLVAR, [11](#)

FitMplus, [11](#)

MplusInput, [19](#)

* **compress**

Compress, [2](#)

* **data**

params, [21](#)

results, [21](#)

* **figure**

FigBias, [4](#)

FigCoverage, [5](#)

FigPower, [6](#)

FigRMSE, [7](#)

FigType1Error, [8](#)

* **fit**

FitDTVAR, [9](#)

FitMLVAR, [11](#)

FitMplus, [11](#)

MplusInput, [19](#)

SimFitDTVAR, [24](#)

SimFitMetaVAR, [25](#)

SimFitMLVAR, [26](#)

SimFitMplus, [27](#)

* **gendata**

Dynamics, [3](#)

GenData, [13](#)

SimGenData, [28](#)

* **manCTMed**

SimFN, [28](#)

* **manMetaVAR**

Compress, [2](#)

Dynamics, [3](#)

FigBias, [4](#)

FigCoverage, [5](#)

FigPower, [6](#)

FigRMSE, [7](#)

FigType1Error, [8](#)

FitDTVAR, [9](#)

FitMetaVAR, [10](#)

FitMLVAR, [11](#)

FitMplus, [11](#)

GenData, [13](#)

MplusInput, [19](#)

Sim, [22](#)

SimFitDTVAR, [24](#)

SimFitMetaVAR, [25](#)

SimFitMLVAR, [26](#)

SimFitMplus, [27](#)

SimGenData, [28](#)

SimProj, [29](#)

Sum, [30](#)

SumFitMetaVARLB, [31](#)

SumFitMetaVARNormal, [31](#)

SumFitMLVAR, [32](#)

SumFitMplus, [33](#)

* **meta**

FitMetaVAR, [10](#)

* **methods**

manmetavar-data-methods, [13](#)

manmetavar-dtvar-methods, [14](#)

manmetavar-metavar-methods, [16](#)

manmetavar-mlvar-methods, [17](#)

manmetavar-mplus-methods, [18](#)

* **parameters**

- params, 21
- * **results**
 - results, 21
- * **simulation**
 - Sim, 22
 - SimFitDTVAR, 24
 - SimFitMetaVAR, 25
 - SimFitMLVAR, 26
 - SimFitMplus, 27
 - SimFN, 28
 - SimGenData, 28
 - SimProj, 29
 - Sum, 30
 - SumFitMetaVARLB, 31
 - SumFitMetaVARNormal, 31
 - SumFitMLVAR, 32
 - SumFitMplus, 33
- * **summary**
 - Sum, 30
 - SumFitMetaVARLB, 31
 - SumFitMetaVARNormal, 31
 - SumFitMLVAR, 32
 - SumFitMplus, 33
- coef.manmetavar.dtvvar
 - (manmetavar-dtvvar-methods), 14
- coef.manmetavar.metavar
 - (manmetavar-metavar-methods), 16
- coef.manmetavar.mplus
 - (manmetavar-mplus-methods), 18
- Compress, 2
- confint.manmetavar.metavar
 - (manmetavar-metavar-methods), 16
- confint.manmetavar.mplus
 - (manmetavar-mplus-methods), 18
- Dynamics, 3, 13
- FigBias, 4, 5–8
- FigCoverage, 4, 5, 6–8
- FigPower, 4, 5, 6, 7, 8
- FigRMSE, 4–6, 7, 8
- FigTypeError, 4–7, 8
- FitDTVAR, 9, 10–12, 20
- FitDTVAR(), 10
- fitDTVARmID::fitDTVARmID, 9
- FitMetaVAR, 9, 10, 11, 12, 20
- FitMLVAR, 9, 10, 11, 12, 20
- FitMplus, 9, 10, 11, 11, 20
- GenData, 3, 13
- GenData(), 9–12
- manmetavar-data-methods, 13
- manmetavar-dtvvar-methods, 14
- manmetavar-metavar-methods, 16
- manmetavar-mlvar-methods, 17
- manmetavar-mplus-methods, 18
- metaVAR::metaVAR, 10
- mlVAR::mlVAR, 11
- MplusInput, 9–12, 19
- params, 21
- plot.manmetavar.data
 - (manmetavar-data-methods), 13
- plot.manmetavar.mplus
 - (manmetavar-mplus-methods), 18
- print.manmetavar.data
 - (manmetavar-data-methods), 13
- print.manmetavar.dtvvar
 - (manmetavar-dtvvar-methods), 14
- print.manmetavar.metavar
 - (manmetavar-metavar-methods), 16
- print.manmetavar.mlvar
 - (manmetavar-mlvar-methods), 17
- results, 21
- Sim, 22
- SimFitDTVAR, 24
- SimFitMetaVAR, 25
- SimFitMLVAR, 26
- SimFitMplus, 27
- SimFN, 28
- SimGenData, 28
- SimProj, 29
- simStateSpace::SimSSMIVary(), 13
- Sum, 30
- SumFitMetaVARLB, 31
- SumFitMetaVARNormal, 31
- SumFitMLVAR, 32
- SumFitMplus, 33
- summary.manmetavar.data
 - (manmetavar-data-methods), 13
- summary.manmetavar.dtvvar
 - (manmetavar-dtvvar-methods), 14

`summary.manmetavar.metavar`
 (`manmetavar-metavar-methods`),
 16

`summary.manmetavar.mlvar`
 (`manmetavar-mlvar-methods`), 17

`summary.manmetavar.mplus`
 (`manmetavar-mplus-methods`), 18

`vcov.manmetavar.dtvvar`
 (`manmetavar-dtvvar-methods`), 14

`vcov.manmetavar.metavar`
 (`manmetavar-metavar-methods`),
 16

`vcov.manmetavar.mplus`
 (`manmetavar-mplus-methods`), 18